Group-3 Submission

Members - Task:

1. Athiyo Chakma (2022118) - Q3
2. Dev Utkarsh Pal (2022150) - Q4
3. Kunjkumar Prajapati (PhD25206) - Q1
4. GARGI (MT25208) - Q2

GitHub Link: https://github.com/deecod-dev/comeg.git ReadMe File

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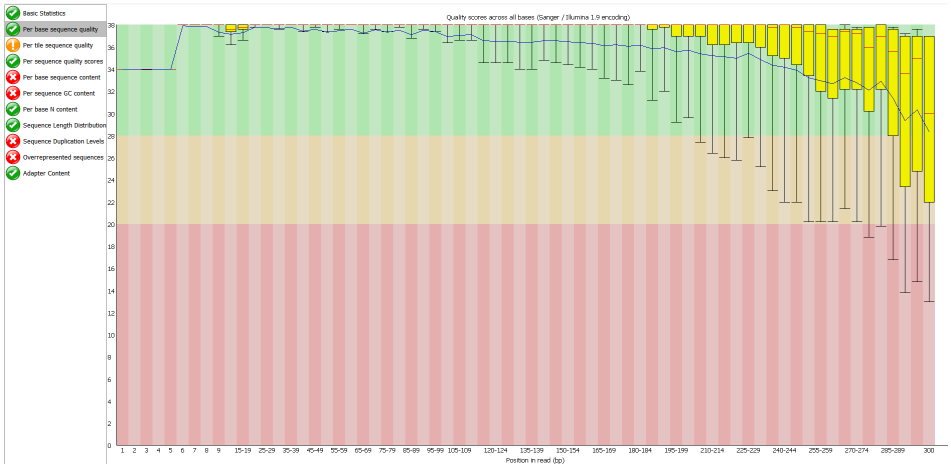
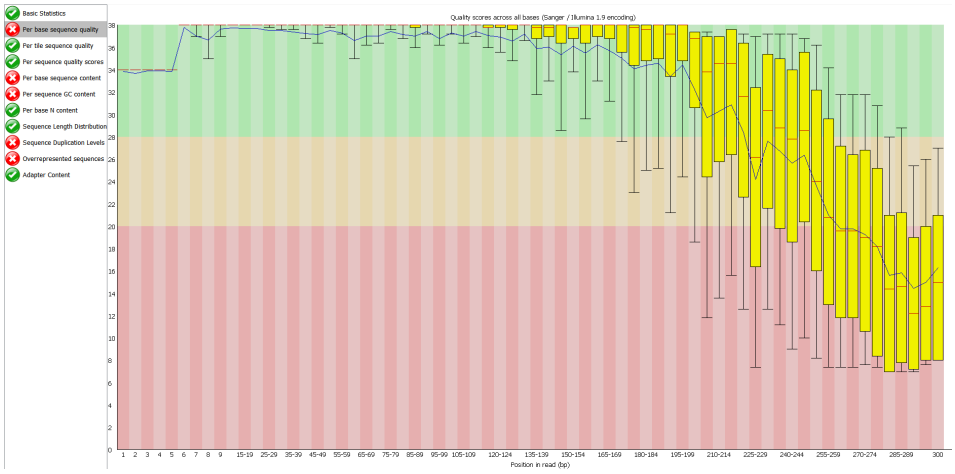
Q1- Quality Check:

Sample-Specific Observations:



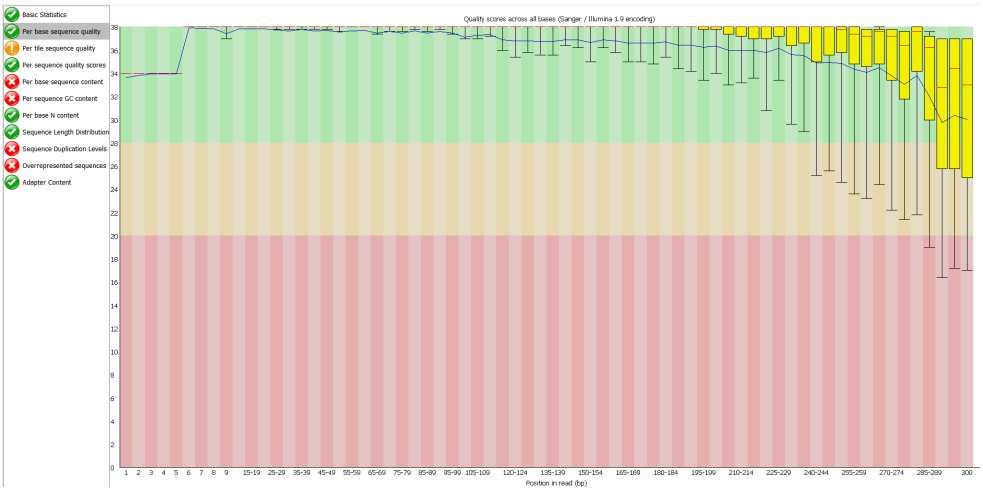
ERR4042824_1



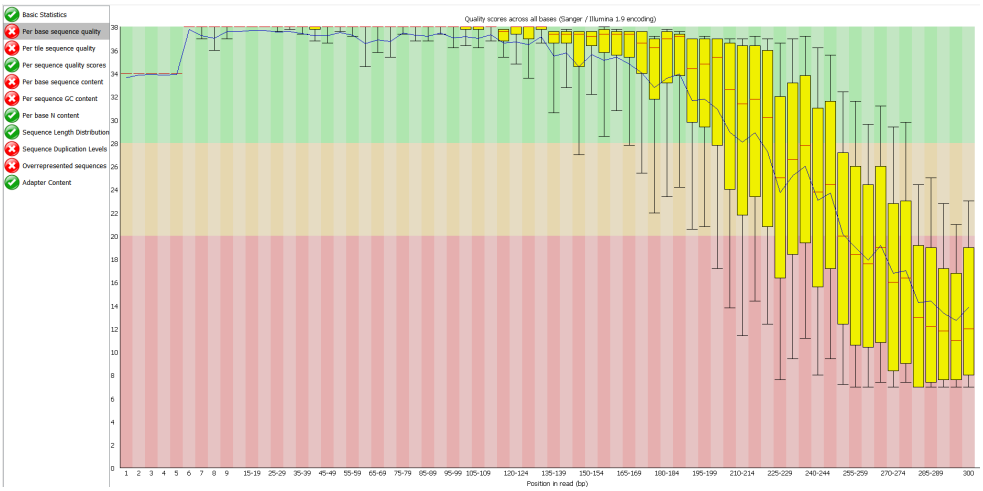
ERR4042824_2

Sample ERR4042824

- Forward Read (_1): High and stable quality through most of the read length, with only a mild decline toward the end.
- Reverse Read (_2): Noticeable and sharper quality drop after ~180 bp, reaching poor quality by the end.
    - *Reverse reads are less reliable and need trimming.*



ERR4042849_1



ERR4042849_2

Sample ERR4042849

- Forward Read (_1): Maintained excellent quality across nearly all positions, minor decline near the final 30 -40 bases.

- Reverse Read (_2): Marked quality deterioration after ~160 bp, falling deep into the poor-quality zone by read endcla
    - *Strong evidence for removing low-quality tails.*

Overview:

FastQC analysis was performed on all raw paired-end reads to assess the sequencing quality before any preprocessing. The "Per Base Sequence Quality" plots were inspected for both the forward (_1) and reverse (_2) reads across multiple samples.

General Trends Across All Samples:

- High Quality at the Start:
  The first ~180 -200 base pairs showed excellent sequencing quality. Median Phred scores were consistently above 28 (green region), indicating highly reliable base calls.

- Progressive Drop in Quality Toward 3′ End:
  Beyond ~180 bp, a consistent decline in quality was observed, particularly in the reverse reads. The median scores dropped into the orange ("acceptable") and red ("poor") zones, signifying increased error probability.

- Higher Variability at Read Ends:
  The expanding boxplots and whiskers toward the end of reads reflect greater variation in base quality; some reads remain good, while others degrade significantly.

Interpretation & Conclusion:

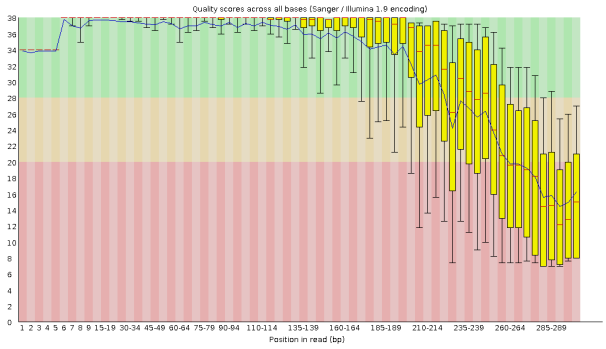
- Forward reads are of consistently high quality.
- Reverse reads suffer from significant degradation near their 3′ ends.
- This decline is typical for Illumina data and can introduce sequencing errors if left untrimmed.

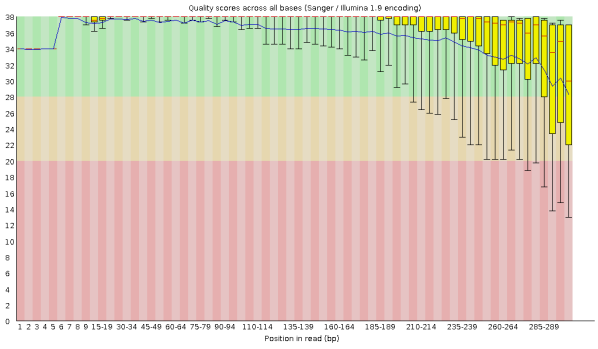
These findings **justify the need for quality trimming** using **Trimmomatic**, which removes low-quality regions and ensures that only high-confidence reads are retained for downstream microbiome analysis.

Q2- Trimmomatic:

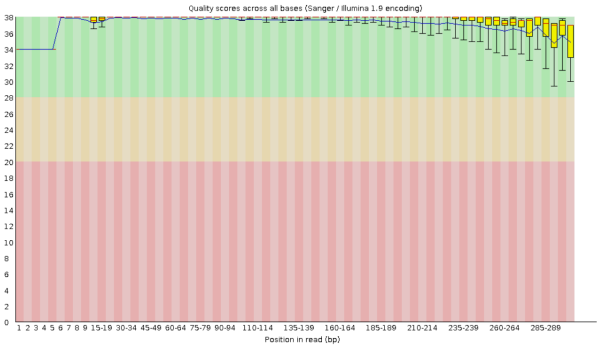
Quality Assessment - Before vs After Trimming:

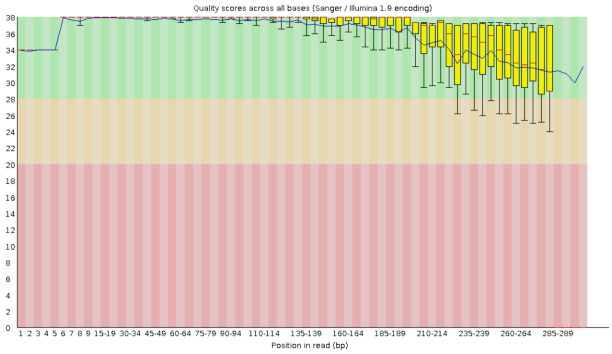
Sample ERR4042824 before Trimmomatic:





Sample ERR4042824 after Trimmomatic:





Methods:

Tool: Trimmomatic v0.39

Parameters Used:

- **SLIDINGWINDOW:5:27** - Scan read with 5-base window, trim when average quality drops below 27
- **MINLEN:100** - Discard reads shorter than 100 bp after trimming
- **AVGQUAL:27** - Remove reads with average quality below 27
- **ILLUMINACLIP** - Remove Illumina adapter sequences
- **PE** - Paired-end mode to maintain read pairs

Commands:

```
cd ~/Batch3_16S
mkdir -p trimmed
conda activate trimmomatic_env
```

```
# Path to adapter file
ADAPTERS=$(dirname $(which trimmomatic))/../share/trimmomatic/adapters/TruSeq3-PE.fa

for f1 in *_1.fastq.gz; do
   f2=${f1/_1.fastq.gz/_2.fastq.gz}
   base=${f1%_1.fastq.gz}

   trimmomatic PE -threads 4 -phred33 \
      $f1 $f2 \
      trimmed/${base}_1_paired.fastq.gz trimmed/${base}_1_unpaired.fastq.gz \
      trimmed/${base}_2_paired.fastq.gz trimmed/${base}_2_unpaired.fastq.gz \
      ILLUMINACLIP:${ADAPTERS}:2:30:10 \
      SLIDINGWINDOW:5:27 AVGQUAL:27 MINLEN:100
done

# Post-trimming quality check
mkdir fastqc_trimmed
fastqc -t 4 -o fastqc_trimmed trimmed/*_paired.fastq.gz
multiqc -o fastqc_trimmed fastqc_trimmed
```

Results and Output:

Output Files Generated:

- Paired Reads: *_paired.fastq.gz - High-quality reads with both forward and reverse pairs intact
- Unpaired Reads: *unpaired.fastq.gz - Orphaned reads where one mate was discarded
- 20 samples × 2 files each = 40 output files

```
    Directory: C:\Users\athiy\Downloads\Batch3_16S\Batch3_16S\trimmed


Mode                LastWriteTime         Length Name
----                -------------         ------ ----
-a----        06-10-2025     19:58        4627816 ERR4042824_1_paired.fastq.gz
-a----        06-10-2025     19:58         295108 ERR4042824_1_unpaired.fastq.gz
-a----        06-10-2025     19:58        4538353 ERR4042824_2_paired.fastq.gz
-a----        06-10-2025     19:58         130675 ERR4042824_2_unpaired.fastq.gz
-a----        06-10-2025     19:58        3382567 ERR4042849_1_paired.fastq.gz
-a----        06-10-2025     19:58         307622 ERR4042849_1_unpaired.fastq.gz
-a----        06-10-2025     19:58        3523143 ERR4042849_2_paired.fastq.gz
-a----        06-10-2025     19:58          68522 ERR4042849_2_unpaired.fastq.gz
-a----        06-10-2025     19:58        2749123 ERR4042883_1_paired.fastq.gz
-a----        06-10-2025     19:58         180610 ERR4042883_1_unpaired.fastq.gz
-a----        06-10-2025     19:58        2739611 ERR4042883_2_paired.fastq.gz
-a----        06-10-2025     19:58         114549 ERR4042883_2_unpaired.fastq.gz
-a----        06-10-2025     19:58        4683927 ERR4042921_1_paired.fastq.gz
-a----        06-10-2025     19:58         279067 ERR4042921_1_unpaired.fastq.gz
-a----        06-10-2025     19:58        4715203 ERR4042921_2_paired.fastq.gz
-a----        06-10-2025     19:58         168718 ERR4042921_2_unpaired.fastq.gz
-a----        06-10-2025     19:58        6299885 ERR4042941_1_paired.fastq.gz
-a----        06-10-2025     19:58         436732 ERR4042941_1_unpaired.fastq.gz
-a----        06-10-2025     19:58        6454876 ERR4042941_2_paired.fastq.gz
-a----        06-10-2025     19:58         204231 ERR4042941_2_unpaired.fastq.gz
-a----        06-10-2025     19:58        5390466 ERR4042949_1_paired.fastq.gz
-a----        06-10-2025     19:58          97206 ERR4042949_1_unpaired.fastq.gz
-a----        06-10-2025     19:58        5582587 ERR4042949_2_paired.fastq.gz
-a----        06-10-2025     19:58         287792 ERR4042949_2_unpaired.fastq.gz
-a----        06-10-2025     19:58        2995827 ERR4043086_1_paired.fastq.gz
-a----        06-10-2025     19:58          27164 ERR4043086_1_unpaired.fastq.gz
-a----        06-10-2025     19:58        2941823 ERR4043086_2_paired.fastq.gz
-a----        06-10-2025     19:58          26623 ERR4043086_2_unpaired.fastq.gz
-a----        06-10-2025     19:58        4151235 ERR4043188_1_paired.fastq.gz
-a----        06-10-2025     19:58          84024 ERR4043188_1_unpaired.fastq.gz
-a----        06-10-2025     19:58        4376509 ERR4043188_2_paired.fastq.gz
-a----        06-10-2025     19:58          73074 ERR4043188_2_unpaired.fastq.gz
-a----        06-10-2025     19:58        3203119 ERR4043295_1_paired.fastq.gz
-a----        06-10-2025     19:58         321936 ERR4043295_1_unpaired.fastq.gz
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-a----        06-10-2025     19:58          87949 ERR4043299_2_unpaired.fastq.gz
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-a----        06-10-2025     19:58          28123 ERR4043596_2_unpaired.fastq.gz
-a----        06-10-2025     19:58        3153769 ERR4043606_1_paired.fastq.gz
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-a----        06-10-2025     19:58          27142 ERR4043606_2_unpaired.fastq.gz
-a----        06-10-2025     19:58        3816257 ERR4043608_1_paired.fastq.gz
-a----        06-10-2025     19:58          32025 ERR4043608_1_unpaired.fastq.gz
-a----        06-10-2025     19:58        3633115 ERR4043608_2_paired.fastq.gz
-a----        06-10-2025     19:58          28267 ERR4043608_2_unpaired.fastq.gz
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-a----        06-10-2025     19:58        4287219 ERR4043615_2_paired.fastq.gz
-a----        06-10-2025     19:58          34414 ERR4043615_2_unpaired.fastq.gz
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-a----        06-10-2025     19:58          92286 ERR4043792_2_unpaired.fastq.gz
-a----        06-10-2025     19:58        3665258 ERR4043803_1_paired.fastq.gz
-a----        06-10-2025     19:58         346743 ERR4043803_1_unpaired.fastq.gz
-a----        06-10-2025     19:58        3729519 ERR4043803_2_paired.fastq.gz
-a----        06-10-2025     19:58         117016 ERR4043803_2_unpaired.fastq.gz
-a----        06-10-2025     19:58        3674666 ERR4043806_1_paired.fastq.gz
-a----        06-10-2025     19:58         370220 ERR4043806_1_unpaired.fastq.gz
-a----        06-10-2025     19:58        3817661 ERR4043806_2_paired.fastq.gz
-a----        06-10-2025     19:58         100337 ERR4043806_2_unpaired.fastq.gz
```

Overall Statistics from MultiQC:

Total Samples Processed: 20 (40 FASTQ files)

Total Sequences After Trimming: ~1.1 million reads

Average Read Retention: ~85% (estimated from sequence counts)

GC Content Range: 50-55% (typical for 16S rRNA)

Duplicate Rate: 3-7% (acceptable for amplicon sequencing)

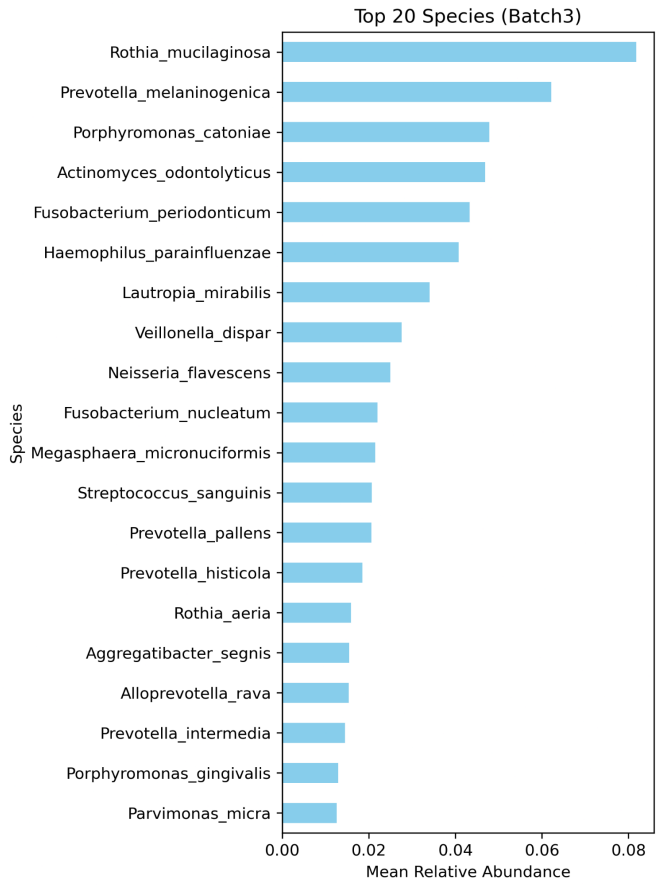
Q3- SPINGO:

Species Countable:
https://github.com/deecod-dev/comeg/blob/main/Batch3_16S/species_counts_table.tsv

Species Relative Abundance:
https://github.com/deecod-dev/comeg/blob/main/Batch3_16S/species_rel_abundance.tsv

Top 20 Species Mean Abundance:
https://github.com/deecod-dev/comeg/blob/main/Batch3_16S/top20_species_mean_abundance.tsv

Top-20 Species Bar Plot (sorted):



Commands Used:

```
#!/bin/bash
# Q-3 SPINGO pipeline for Batch3

# Paths
SPINGO_PATH=~/Batch3_16S/SPINGO-master/spingo
RDP_DB=~/Batch3_16S/SPINGO-master/database/RDP_database/database/RDP_11.2.species.fa
THREADS=10
```

```
# Directories
mkdir -p spingo_fasta spingo_out

echo "Step 1: Running SPINGO on all paired samples..."

# Loop through all paired trimmed samples
for f1 in trimmed/*_1_paired.fastq.gz; do
    base=$(basename $f1 _1_paired.fastq.gz)
    f2=trimmed/${base}_2_paired.fastq.gz

    echo "Processing sample: $base"

    # concatenate paired reads
    zcat $f1 $f2 > spingo_fasta/${base}.fastq

    # convert fastq -> fasta
    awk 'NR%4==1 {gsub("@",">",$0); print} NR%4==2 {print}' spingo_fasta/${base}.fastq > spingo_fasta/${base}.fasta

    # run SPINGO
    $SPINGO_PATH -d $RDP_DB -p $THREADS -i spingo_fasta/${base}.fasta > spingo_out/${base}.spingo.out.txt

    # clean intermediates
    rm spingo_fasta/${base}.fastq spingo_fasta/${base}.fasta
done

echo "Step 2: Parsing SPINGO outputs to create counts table..."

# Python inline for parsing and plotting
python3 << 'EOF'
import glob, pandas as pd
import matplotlib.pyplot as plt

files = sorted(glob.glob('spingo_out/*.spingo.out.txt'))
all_species = {}

for fn in files:
    sample = fn.split('/')[-1].replace('.spingo.out.txt','')
    counts = {}
    with open(fn) as fh:
        for line in fh:
            if line.startswith('#'): continue
            cols = line.strip().split('\t')
            species, species_conf = cols[-2], float(cols[-1])
            # filter by confidence >= 0.7 and remove ambiguous
            if species_conf >= 0.7 and species.lower() != 'ambiguous':
                counts[species] = counts.get(species,0)+1
    all_species[sample] = counts


# Create counts dataframe
species_list = sorted({s for d in all_species.values() for s in d.keys()})
df = pd.DataFrame(0, index=species_list, columns=all_species.keys())
for samp, d in all_species.items():
    for sp, c in d.items():
        df.at[sp, samp] = c

df.to_csv('species_counts_table.tsv', sep='\t')
print("species_counts_table.tsv created.")

# Total-sum scaling normalization
```

```
rel_ab = df.div(df.sum(axis=0), axis=1)
rel_ab.to_csv('species_rel_abundance.tsv', sep='\t')
print("species_rel_abundance.tsv created.")

# Top 20 species by mean relative abundance
top20 = rel_ab.mean(axis=1).sort_values(ascending=False).head(20)
top20.to_csv('top20_species_mean_abundance.tsv', sep='\t', header=['mean_abundance'])

# Plot top 20 species
top20.sort_values().plot(kind='barh', figsize=(6,8), color='skyblue')
plt.xlabel('Mean Relative Abundance')
plt.ylabel('Species')
plt.title('Top 20 Species (Batch3)')
plt.tight_layout()
plt.savefig('top20_species_barplot.png', dpi=300)
# plt.show()
EOF

echo "Q-3 pipeline finished. Outputs:"
echo "- species_counts_table.tsv"
echo "- species_rel_abundance.tsv"
echo "- top20_species_mean_abundance.tsv"
echo "- top20_species_barplot.png"
```

Computation Parameters:

- Threads: 10
- k-mer size: 8
- Bootstrap samples: 10
- Subsample size: 1/8

Each sample was matched against ~95 k reference 16S sequences.

Q4- Diversity Analysis (Shannon & Pielou):

Diversity Indices Formula:

Shannon Index (H')
$H = -\sum(p_i \times \ln(p_i))$

Pielou's Evenness (J)
$J = H / \ln(S)$
Where S = number of species in sample

Statistical Analysis:

Tools: R with vegan package
Group comparisons: Study condition (2 groups) and age category (3 groups)
Visualization: Boxplots with statistical annotation

Statistical Summary:

By Study Condition:
Shannon Diversity: Control (3.70) vs Diseased (3.60) - Δ = 0.10
Pielou's Evenness: Control (0.71) vs Diseased (0.72) - Δ = 0.01

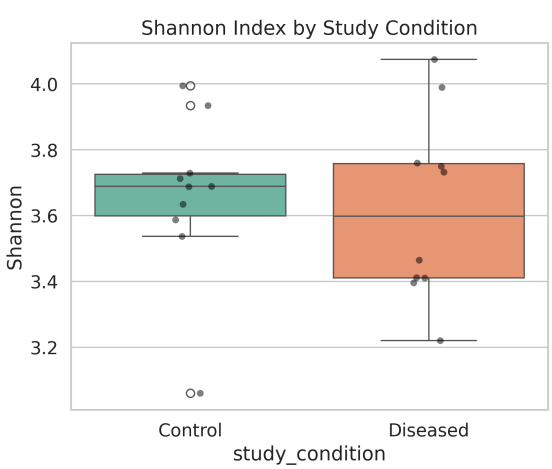
By Age Category:
Shannon Diversity: Young (3.72) → Middle (3.60) → Senior (3.71)
Pielou's Evenness: Young (0.725) → Middle (0.695) → Senior (0.70)

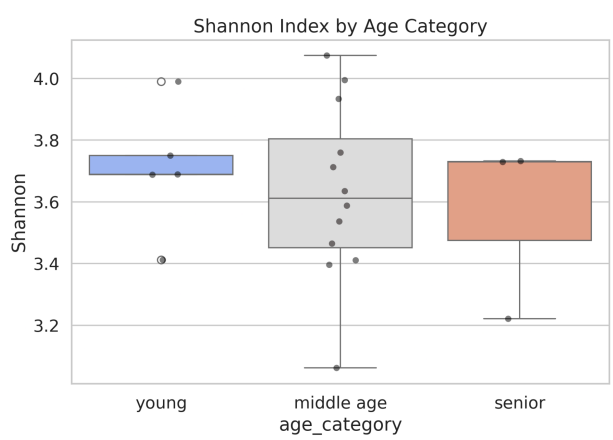
Results:

Diversity Summary Indices Table:
https://github.com/deecod-dev/comeg/blob/main/Batch3_16S/diversity_indices.tsv

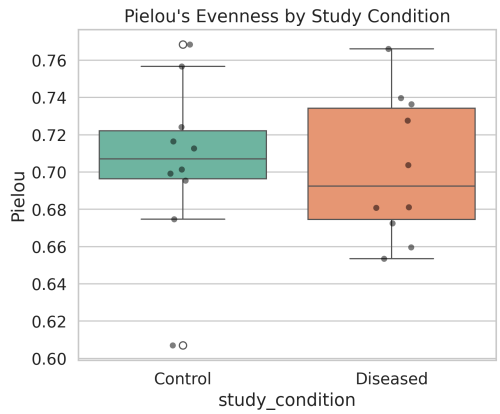
Visualization:

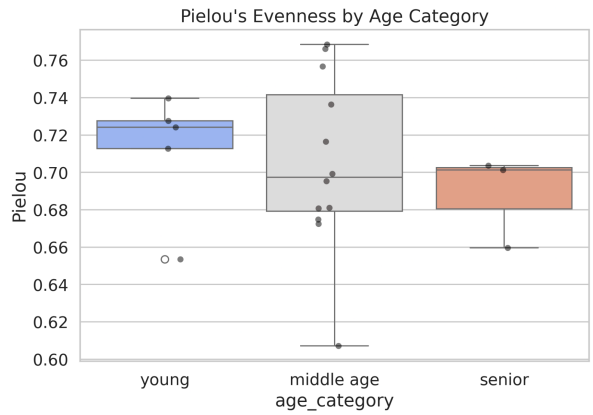
Shannon:





Pileou:





Interpretations:

1.Shannon Index by Study Condition

- Control: Median = 3.70
- Diseased: Median = 3.6

Interpretation:

- Control samples show slightly higher Shannon diversity than Diseased samples
- The small difference (0.10 Shannon units) suggests disease state has minimal impact on microbial diversity
- This may indicate the disease condition does not substantially alter overall microbiome richness

2.Shannon Index by Age Category

- Young: Median = 3.72
- Middle: Median = 3.60
- Senior: Median = 3.71

Interpretation:

- Unexpected U-shaped pattern with middle age showing the lowest diversity
- Young and senior individuals maintain comparable diversity levels
- Middle age may represent a period of particular microbiome instability

3.Pielou's Evenness by Study Condition

- Control: Median = 0.71
- Diseased: Median = 0.72

Interpretation:

- Virtually similar evenness between Control and Diseased groups

- Species distribution patterns remain stable regardless of health status
- Suggests disease condition does not disrupt the balance among microbial species

4.Pielou's Evenness by Age Category

- Young: Median = 0.725
- Middle: Median = 0.695
- Senior: Median = 0.70

Interpretation:

- Middle-aged individuals show reduced species evenness compared to other age groups
- Young individuals maintain the most balanced microbial communities
- Seniors demonstrate partial recovery of species balance compared to middle age